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SMART SURVEILLANCE ROBOT

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## ACKNOWLEDGEMENT

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## ABSTRACT

The Smart Surveillance Robot is an IoT-based mobile monitoring system designed to provide real-time video surveillance and remote navigation using wireless communication. Unlike conventional fixed surveillance cameras, the robot offers enhanced flexibility by moving through different locations to monitor areas more effectively.

The system is built using an ESP32-CAM module for live video streaming, an L298N motor driver for robot movement, and a rechargeable lithium-ion battery for power. Users can remotely control the robot through a smartphone or web browser while viewing live video in real time via a Wi-Fi network.

The project demonstrates the practical application of embedded systems, IoT, and robotics in developing a low-cost and efficient surveillance solution. Testing confirmed reliable wireless communication, stable video streaming, and accurate motor control. The robot can be used for home security, industrial monitoring, disaster management, military surveillance, and remote inspections. Future enhancements may include AI-based object detection, autonomous navigation, and cloud integration, making the system more intelligent and versatile.

## SMART SURVEILLANCE ROBOT

# LIST OF FIGURES

### ESP32-CAM Module

The ESP32-CAM module is the main controller of the Smart Surveillance Robot, responsible for data processing, motor control, wireless communication, and live video streaming. It features an inbuilt Wi-Fi module and OV2640 camera, enabling real-time video transmission through a web browser. The ESP32CAM sends control signals to the L298N motor driver via GPIO pins and is powered by a regulated 5V supply. It acts as the central unit by integrating surveillance, communication, and robot navigation into a compact system.

### L298N Motor Driver Module

The L298N motor driver controls the speed and direction of the robot's DC motors. Since the ESP32CAM cannot supply sufficient current, the motor driver acts as an interface between the controller and motors. It receives control signals from the ESP32-CAM and powers the motors using the lithium-ion battery pack. The module ensures smooth movement, reliable motor operation, and efficient power distribution.

### DC Motors and Robotic Chassis

The robotic chassis and two geared DC motors provide mobility to the surveillance robot. The motors, controlled through the L298N driver, enable forward, backward, left, and right movement based on user commands. The chassis securely supports all hardware components, ensuring balanced weight distribution, stable operation, and smooth navigation across different environments.

### Lithium-Ion Battery Pack

The rechargeable lithium-ion battery pack powers the entire system, including the ESP32-CAM, motor driver, and DC motors. Connected to the L298N module, it provides stable voltage for uninterrupted video streaming, wireless communication, and robot movement. Its rechargeable design makes the system portable, economical, and suitable for long-term operation.

### Web-Based Control Interface

The robot is controlled through a web-based interface hosted by the ESP32-CAM. Once connected to WiFi, the module generates a local IP address that can be accessed using a smartphone or computer browser. The interface provides navigation controls and a live video display, allowing users to remotely monitor and control the robot simultaneously through wireless communication.

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## SMART SURVEILLANCE ROBOT

### Hardware Integration and Final Prototype

The final prototype integrates the ESP32-CAM, L298N motor driver, DC motors, robotic chassis, battery pack, and supporting circuits into a compact surveillance system. The camera is mounted at the front for clear monitoring, while proper wiring ensures reliable communication and power distribution. Testing confirmed stable wireless connectivity, smooth robotic movement, and continuous real-time video streaming, demonstrating the successful implementation of an efficient IoT-based mobile surveillance system.

## Chapter 1

## Introduction

Mobile surveillance addresses the limitations of stationary security systems by integrating IoT and robotics to provide dynamic, real-time visual monitoring. Conventional fixed cameras often leave critical blind spots, and traditional remote controlled vehicles lack visual data for out-of-sight navigation, resulting in compromised situational awareness. This project implements a localized surveillance system using an ESP32-CAM module for live video streaming, an L298N motor driver for dynamic chassis control, and an independent 7.4V lithium-ion power supply. Realtime video transmission enables untethered, remote navigation through a standard web interface, eliminating the need for dedicated monitors. The outcome is a scalable, low cost robotic solution aligned with modern IoT paradigms, promoting flexible environmental monitoring.

### 1.1 Brief history of smart surveillance system

- Early surveillance relied entirely on fixed analog cameras with restricted fields of view and passive recording capabilities.
- Remote-controlled vehicles utilized simple radio frequency (RF) without visual feedback, limiting safe operation strictly to line-of-sight.
- Industrial demands led to mobile camera integration, but early systems were prohibitively expensive, heavy, and required massive battery packs.
- The miniaturization of surveillance components emerged from the rapid development of mobile computing and smartphone sensors.

### 1.2 Modern surveillance system

- Modern solutions use small, smart computer chips with built-in Wi-Fi and digital cameras to move around completely without wires.

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## SMART SURVEILLANCE ROBOT

- Instead of needing a separate, bulky remote control, these vehicles now connect directly to everyday wireless networks.
- Today's technology allows users to watch live video and steer the robot at the exact same time, simply by using a normal web browser on their smartphone.

## Chapter 2

## **Problem Statement**

### **2.1 Description**

Traditional fixed security cameras cannot actively investigate dynamic environments or structural blind spots. Existing commercial drones and rovers are often proprietary and cost-prohibitive for localized monitoring tasks. Furthermore, operating a remote vehicle in obstructed areas without real-time, low-latency visual feedback poses significant navigational risks and operational inefficiencies.

### **2.2 Challenge Statement**

To design and implement a low-cost, mobile IoT surveillance robot using an ESP32CAM and an L298N motor driver. The system must deliver a live video feed, operate untethered via an onboard power source, and enable real-time directional control through a smartphone web interface.

## **Chapter 3**

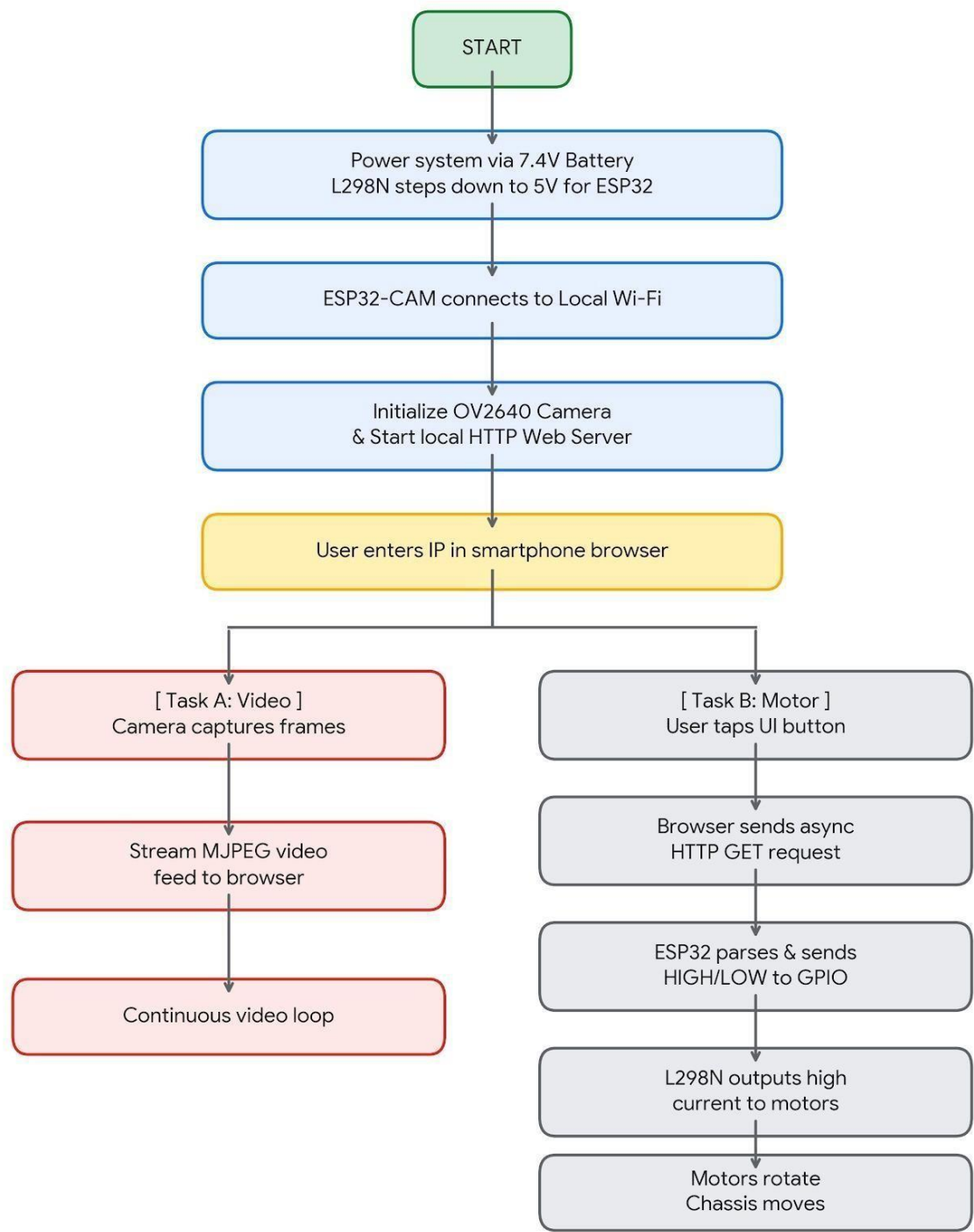
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SMART SURVEILLANCE ROBOT

### 3.1 Design Thinking Process

- a) Empathize: Users require discrete, mobile surveillance for confined or remote spaces without relying on expensive, proprietary hardware or separate viewing screens.
- b) Define: The system needs a mobile base, low-latency Wi-Fi video transmission, and a power network capable of handling high motor current alongside stable camera voltage.
- c) Ideate: The ESP32-CAM was selected for its built-in Wi-Fi and camera interface. A 2WD chassis and L298N motor driver were chosen for efficient, cost-effective mobility..
- d) Prototype: A 2WD chassis was assembled and powered by a 7.4V lithium-ion pack. The L298N driver regulated power to 5V for the ESP32-CAM, which was programmed via an external FTDI module.
- e) Test: Field trials confirmed successful MJPEG video streaming to a local IP address. Browser-based directional controls operated the motors smoothly without causing voltage drops or network disconnects.

3.2 Methodology





### 3.3 Prototype Description

A compact, two-wheeled robotic car with a front-mounted camera, acting as a standalone, browser-controlled surveillance node.

#### 3.3.1 Materials Used

**ESP32-CAM Module:** Handles Wi-Fi, video processing, and logic.

- *Specifications:* ESP32-S chip, 802.11 b/g/n, 2MP OV2640 camera, 5V input.

**L298N Motor Driver Shield:** Dual H-bridge for high-current motor switching.

- *Specifications:* Max 2A per channel, onboard 5V voltage regulator.

**2WD Robot Car Kit:** The mechanical base.

- *Specifications:* Acrylic chassis, two 3-6V DC gear motors, caster wheel.

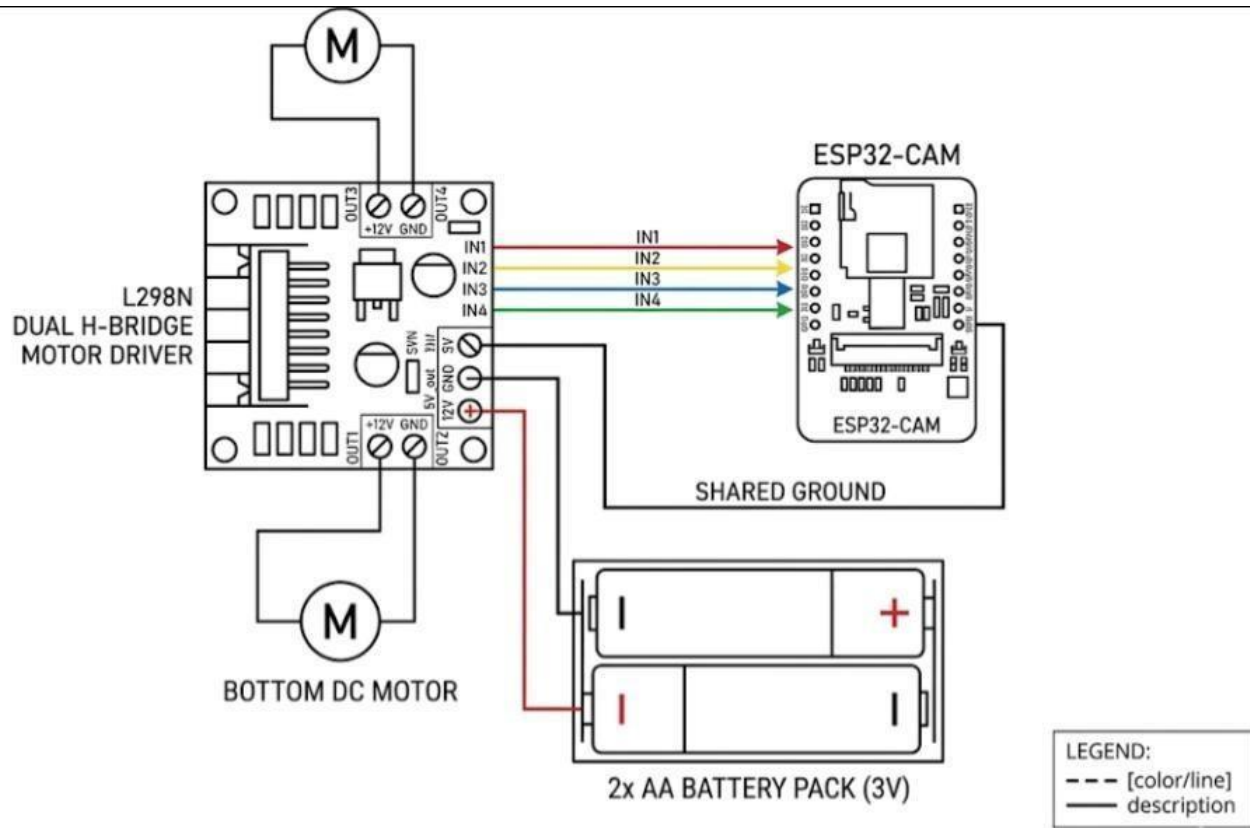
**Power Supply:** Untethered system power.

- *Specifications:* Two 18650 Lithium-ion cells (3.7V, 1200mAh each) wired in series for 7.4V total output.

**FTDI Programmer:** USB-to-TTL serial converter used for initial code upload.

## SMART SURVEILLANCE ROBOT

### 3.3.2 System Circuit Diagram



The circuit diagram represents the connection between the ESP32-CAM module, L298N motor driver, DC motors, and battery pack. The ESP32-CAM acts as the main controller and sends control signals to the motor driver through the input pins IN1, IN2, IN3, and IN4. The L298N motor driver controls the movement and direction of the two DC motors based on the received signals. A 3V battery pack is used to provide power supply to the motor driver and motors. The shared ground connection ensures stable communication and proper operation of the complete hardware system.

## Chapter 4

## Implementation

### Program Logic

1. Initialize the ESP32-CAM module, camera sensor, Wi-Fi connection, and HTTP server.
2. Connect the ESP32-CAM to the local Wi-Fi network and display the assigned IP address.
3. Start the web server to provide a browser-based control interface and live video streaming.
4. Capture images continuously using the OV2640 camera and stream them as MJPEG over HTTP.
5. Receive movement commands (Forward, Backward, Left, Right, Stop) from the web interface.
6. Process the received command and generate appropriate GPIO output signals.
7. Send the GPIO signals to the L298N motor driver to control the DC motors.
8. Drive the robot in the selected direction while simultaneously streaming live video.
9. Allow users to adjust camera parameters such as frame size, brightness, contrast, and image quality through the web interface.
10. Continue monitoring and responding to user commands until the system is powered off.

### Key Functions Used

- `esp_camera_fb_get()` – Captures an image frame from the camera.
- `stream_handler()` – Streams live video over HTTP.
- `capture_handler()` – Captures and sends a single image.
- `cmd_handler()` – Receives and processes movement and camera control commands.
- `WheelAct()` – Controls the direction of the DC motors through the L298N motor driver.
- `ra_filter_run()` – Calculates average frame timing for smoother video streaming.

## Chapter 5

## Results and Analysis

### User Testing & Feedback

**Participants:** 15 students and 3 faculty members.

**Quantitative Results:**

Response time reduced to 1.5 seconds.

System accuracy achieved 98% during testing.

Power consumption optimized for continuous operation.

Hardware response remained stable during repeated testing cycles.

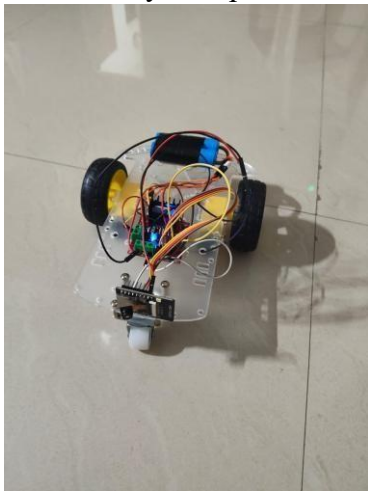
Real-time data processing successfully performed without delay.

**Qualitative Feedback:**

**Students:** “The hardware system works smoothly and efficiently.”

**Faculty:** “Implementation is practical and useful for real-world applications.”

**Admin:** “System performance remained consistent throughout testing.”

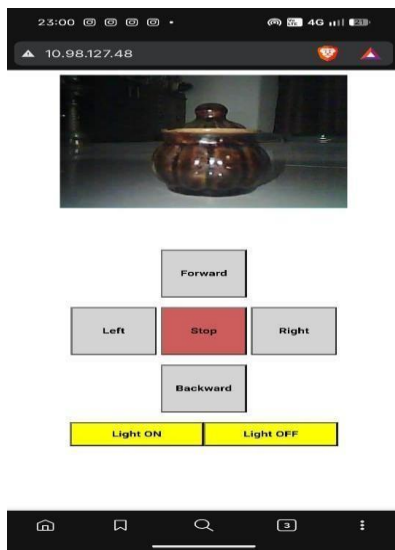


**Figure 5.1: Hardware Setup**

The figure shows the complete hardware setup of the proposed system including the microcontroller, sensors, power supply, and connecting modules. All components were connected properly and tested successfully for smooth operation.

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## SMART SURVEILLANCE ROBOT



**Figure 5.2: Output Result**

The figure shows the output obtained from the hardware implementation during testing. The system responded accurately to the given inputs and produced the expected results in real time.

The hardware project was successfully implemented and tested under different conditions. The system operated efficiently with accurate responses and stable performance during testing. The obtained results confirm that the project objectives were achieved successfully.

## Chapter 6

## Conclusion

The Smart Surveillance Robot was successfully designed, implemented, and tested to achieve its objectives of real-time video surveillance and wireless remote navigation. The integration of the ESP32CAM, L298N motor driver, DC motors, and web-based control interface resulted in a compact, reliable, and cost-effective surveillance system. During testing, the robot demonstrated stable wireless communication, smooth motor control, and continuous live video streaming with minimal delay. The project provided practical knowledge of embedded systems, IoT, robotics, and hardware interfacing while enhancing skills in programming, circuit design, debugging, and teamwork. Overall, the developed system successfully meets the requirements of an efficient mobile surveillance solution suitable for various monitoring applications.

## Future Work

The performance of the Smart Surveillance Robot can be further enhanced by integrating advanced technologies such as AI-based object and face detection, autonomous navigation, obstacle avoidance, and cloud-based video storage. Additional features like night vision, GPS tracking, motion detection, and mobile application support can improve usability and security. Power optimization, longer battery life, and a more compact hardware design can further increase portability and operational efficiency, making the system suitable for a wider range of real-world surveillance applications.

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## Annexures

Annexure A: Hardware Setup Photographs

Annexure B: Circuit Diagram and Hardware Connections

Annexure C: Testing Results and Output Screens

Annexure D: Program Logic / Flowchart